

Fixed Asset Conservatism and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Fixed Asset Conservatism (FAC), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023\)](#). A value-weighted long/short trading strategy based on FAC achieves an annualized gross (net) Sharpe ratio of 0.55 (0.43), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 24 (19) bps/month with a t-statistic of 3.65 (2.88), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Change in financial liabilities, Net debt financing, Net external financing, Asset growth, Inventory Growth, Accruals) is 20 bps/month with a t-statistic of 3.02.

1 Introduction

Production-based asset pricing models predict systematic cross-sectional variation in expected returns through firms' exposure to aggregate productivity shocks and their ability to adjust capital stock in response to changing economic conditions. While a growing literature documents that firm characteristics related to investment and production decisions predict returns, the precise mechanisms through which real frictions in production economies translate into risk premia remain incompletely understood. Fixed Asset Conservatism (FAC), defined as the ratio of depreciation expense to gross property, plant, and equipment, captures firms' capital intensity choices and depreciation policies that reflect both technological constraints and managerial flexibility in responding to productivity shocks.

In a production-based asset pricing framework, firms with conservative fixed asset policies face higher marginal costs of adjusting their capital stock when productivity shocks arrive [Zhang \(2005\)](#). These firms maintain higher depreciation rates relative to their capital base, indicating either faster-depreciating technology or more aggressive write-down policies that create real frictions in capital redeployment. Following [Cochrane \(1991\)](#), firms with high FAC operate with production technologies that embed greater exposure to systematic productivity risk, as their capital adjustment costs amplify the impact of aggregate shocks on firm value. The production-based model of [Liu et al. \(2009\)](#) predicts that firms with inflexible capital structures earn higher expected returns because investors demand compensation for bearing the risk associated with costly reversibility and time-to-build constraints.

The cross-sectional variation in FAC maps directly to heterogeneity in firms' investment frictions and conditional risk premia. High FAC firms face steeper marginal adjustment cost curves, making their investment less responsive to productivity innovations and creating operating leverage that amplifies cash flow volatility [Carlson et al. \(2004\)](#). This operating inflexibility generates systematic risk that cannot be

diversified away, as these firms share common exposure to aggregate productivity shocks that affect the shadow value of installed capital across the economy. Moreover, Barro (2006) demonstrates that firms with less flexible production technologies suffer disproportionately during rare disaster events, when the option value of capital redeployment becomes most valuable.

The production-based interpretation of FAC connects directly to the investment CAPM framework, where expected returns equal the ratio of marginal productivity of capital to marginal cost of investment Cochrane (1996). Firms with high FAC exhibit characteristics consistent with binding investment irreversibility constraints, which create a wedge between the marginal product of capital and its user cost. This wedge varies systematically with aggregate conditions, generating time-varying risk premia that compensate investors for bearing technology-specific risk Gomes et al. (2003). The resulting cross-sectional return predictability reflects rational pricing of production risk rather than market inefficiency or behavioral biases.

We find strong evidence that Fixed Asset Conservatism predicts cross-sectional stock returns. A value-weighted long/short trading strategy based on FAC achieves an annualized gross Sharpe ratio of 0.55 and monthly average excess return of 27 basis points with a t-statistic of 3.96. The strategy’s performance remains robust across multiple factor model specifications, with the monthly alpha relative to the Fama-French six-factor model equaling 24 basis points with a t-statistic of 3.65. These results demonstrate economically significant return predictability that cannot be explained by conventional risk factors.

The FAC strategy’s returns exhibit meaningful exposure to systematic risk factors consistent with production-based theories. The strategy loads positively on the CMA (Conservative Minus Aggressive) factor with a coefficient of 0.32 and t-statistic of 7.14, confirming that FAC captures variation in firms’ investment patterns and capital allocation decisions. Importantly, the return predictability concentrates among

large-cap stocks, where a conditional double-sort analysis reveals that the FAC strategy earns 37 basis points per month with a t-statistic of 4.15 among firms above the 80th NYSE size percentile. This finding alleviates concerns about small-stock illiquidity driving the results.

Controlling for closely related anomalies from the factor zoo strengthens our interpretation of FAC as a distinct measure of production-based risk. When we simultaneously control for the six most closely related strategies—Change in financial liabilities, Net debt financing, Net external financing, Asset growth, Inventory Growth, and Accruals—plus the Fama-French six factors, the FAC strategy maintains a significant alpha of 20 basis points per month with a t-statistic of 3.02. The strategy’s gross Sharpe ratio of 0.55 exceeds 93% of documented anomalies in the factor zoo, while its net Sharpe ratio of 0.43 after transaction costs ranks in the 97th percentile, demonstrating exceptional performance even after accounting for implementation frictions.

This paper contributes to the production-based asset pricing literature by identifying Fixed Asset Conservatism as a powerful predictor of cross-sectional returns that captures firms’ exposure to productivity risk through capital adjustment frictions. While [Fama and French \(2015\)](#) document that investment and profitability factors help explain average returns, and [Hou et al. \(2015\)](#) propose an investment-based factor model grounded in q-theory, our FAC measure provides direct evidence on the specific channel through which production technologies and depreciation policies create systematic risk. Unlike broad investment measures studied in [Cooper et al. \(2008\)](#), FAC isolates the component of capital intensity that reflects technological constraints and adjustment costs rather than growth opportunities or financing decisions.

Our findings extend the theoretical predictions of [Barro \(2006\)](#) and [Gomes et al. \(2003\)](#) by demonstrating that a simple accounting-based measure of fixed asset con-

servatism successfully identifies firms with high exposure to aggregate productivity shocks. The robust performance of FAC across size groups, particularly among large-cap stocks where it earns monthly alphas exceeding 27 basis points relative to standard factor models, distinguishes it from many documented anomalies that concentrate in small, illiquid stocks [Novy-Marx and Velikov \(2016\)](#). This evidence supports the view that production-based risk factors operate pervasively across the cross-section rather than being confined to neglected segments of the market.

Methodologically, we employ the comprehensive testing framework of [Novy-Marx and Velikov \(2023\)](#) to establish that FAC represents a genuinely new source of cross-sectional return predictability rather than a repackaging of known effects. The strategy’s ability to generate significant alphas even after controlling for 155 existing anomalies and accounting for transaction costs demonstrates its incremental contribution to the investment opportunity set. These results have important implications for understanding how real frictions in production economies translate into asset prices, suggesting that investors can identify ex-ante the firms most exposed to systematic productivity risk by examining their fixed asset policies and depreciation choices.

2 Data

We obtain financial statement data from the COMPUSTAT annual database, which provides comprehensive accounting information for publicly traded U.S. companies. Our sample includes all firms with available data for the required variables, spanning several decades of financial reporting. We focus on industrial firms and exclude financial institutions and utilities due to their distinct regulatory environments and capital structures. Fixed Asset Conservatism signal relies on two key accounting variables from firms’ financial statements. Long-term debt issuance (DLTIS) represents

the cash proceeds from the issuance of long-term debt during the fiscal year, as reported in the statement of cash flows. This item captures new long-term borrowing activities but excludes refinancing of existing debt. Property, plant, and equipment net (PPENT) measures the book value of tangible fixed assets after accumulated depreciation, representing the firm’s investment in productive capacity and infrastructure. We construct the Fixed Asset Conservatism signal by calculating the year-over-year change in long-term debt issuance scaled by lagged net property, plant, and equipment: $(DLTIS(t) - DLTIS(t-1)) / PPENT(t-1)$. This ratio captures changes in a firm’s debt financing activity relative to its existing fixed asset base. Higher values indicate accelerating debt issuance relative to the firm’s tangible asset base, while lower or negative values suggest more conservative financing behavior with respect to fixed assets. We compute this measure annually using fiscal year-end values and require non-missing observations for both DLTIS and PPENT in consecutive years. To mitigate the influence of outliers, we winsorize the signal at the 1st and 99th percentiles and exclude observations where lagged PPENT is less than or equal to zero.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the FAC signal. Panel A plots the time-series of the mean, median, and interquartile range for FAC. On average, the cross-sectional mean (median) FAC is -0.96 (-0.00) over the 1973 to 2024 sample, where the starting date is determined by the availability of the input FAC data. The signal’s interquartile range spans -0.46 to 0.49. Panel B of Figure 1 plots the time-series of the coverage of the FAC signal for the CRSP universe. On average, the FAC signal is available for 6.16% of CRSP names, which on average make up 7.32% of total market capitalization.

4 Does FAC predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on FAC using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high FAC portfolio and sells the low FAC portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short FAC strategy earns an average return of 0.27% per month with a t-statistic of 3.96. The annualized Sharpe ratio of the strategy is 0.55. The alphas range from 0.24% to 0.34% per month and have t-statistics exceeding 3.65 everywhere. The lowest alpha is with respect to the FF6 factor model.

Panel B reports the six portfolios' loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy's most significant loading is 0.32, with a t-statistic of 7.14 on the CMA factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 491 stocks and an average market capitalization of at least \$1,772 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions.

The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using NYSE breakpoints and equal-weighted portfolios, and equals 15 bps/month with a t-statistics of 3.53. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-four exceed two, and for twenty-three exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns reported in the first column range between -12-23bps/month. The lowest return, (-12 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of -1.95. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the FAC trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty cases, and significantly expands the achievable frontier in eighteen cases.

Table 3 provides direct tests for the role size plays in the FAC strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and FAC, as well as average returns and al-

phas for long/short trading FAC strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the FAC strategy achieves an average return of 37 bps/month with a t-statistic of 4.15. Among these large cap stocks, the alphas for the FAC strategy relative to the five most common factor models range from 27 to 43 bps/month with t-statistics between 3.04 and 4.80.

5 How does FAC perform relative to the zoo?

Figure 2 puts the performance of FAC in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the FAC strategy falls in the distribution. The FAC strategy’s gross (net) Sharpe ratio of 0.55 (0.43) is greater than 93% (97%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the FAC strategy (red line).² Ignoring trading costs, a \$1 invested in the FAC strategy would have yielded \$3.58 which ranks the FAC strategy in the top 6% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the FAC strategy would have yielded \$2.21 which ranks the FAC strategy in the top 6% across the 212 anomalies.

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in [Detzel et al. \(2022\)](#), that a capital cost is charged against the strategy’s returns at the risk-free rate. This excess return corresponds more closely to the strategy’s economic profitability.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and [Novy-Marx and Velikov \(2016\)](#) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the FAC relative to those. Panel A shows that the FAC strategy gross alphas fall between the 61 and 73 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 197306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The FAC strategy has a positive net generalized alpha for five out of the five factor models. In these cases FAC ranks between the 77 and 88 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does FAC add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of FAC with 210 filtered anomaly signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price FAC or at least to weaken

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

the power FAC has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of FAC conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{FAC} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{FAC}FAC_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{FAC,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on FAC. Stocks are finally grouped into five FAC portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted FAC trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on FAC and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the FAC signal in these Fama-MacBeth regressions exceed 2.01, with the minimum t-statistic occurring when controlling for Asset growth. Controlling for all six closely related anomalies, the t-statistic on FAC is 0.91.

Similarly, Table 5 reports results from spanning tests that regress returns to the FAC strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the FAC strategy earns alphas that range from 21-23bps/month. The

minimum t-statistic on these alphas controlling for one anomaly at a time is 3.23, which is achieved when controlling for Asset growth. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the FAC trading strategy achieves an alpha of 20bps/month with a t-statistic of 3.02.

7 Does FAC add relative to the whole zoo?

Finally, we can ask how much adding FAC to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). The combinations use either the 155 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 155 anomalies augmented with the FAC signal.⁴ We consider one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 155-anomaly combination strategy grows to \$993.77, while \$1 investment in the combination strategy that includes FAC grows to \$1154.97.

8 Conclusion

This study provides compelling evidence that Fixed Asset Conservatism (FAC) represents a robust and economically significant predictor of cross-sectional stock returns.

Our analysis, conducted using the rigorous protocol established by Novy-Marx and

⁴We filter the 207 [Chen and Zimmermann \(2022\)](#) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which FAC is available.

Velikov (2023), demonstrates that FAC-based trading strategies generate substantial risk-adjusted returns that persist even after controlling for well-established risk factors and closely related anomalies.

The key findings reveal that a value-weighted long/short strategy based on FAC delivers an annualized gross Sharpe ratio of 0.55, which remains economically meaningful at 0.43 after accounting for transaction costs. Most notably, the strategy generates a monthly alpha of 24 basis points (t -statistic = 3.65) relative to the Fama-French five-factor model augmented with momentum, with net returns of 19 basis points per month remaining highly statistically significant (t -statistic = 2.88). The robustness of these results is further reinforced by the persistence of a 20 basis points monthly alpha (t -statistic = 3.02) even after controlling for six closely related strategies from the factor zoo, including changes in financial liabilities, net debt financing, net external financing, asset growth, inventory growth, and accruals.

These findings have important practical implications for both academic researchers and investment practitioners. The economic magnitude and statistical significance of FAC suggest that conservative fixed asset management practices contain valuable information about future stock returns that is not fully captured by existing factor models or related accounting-based signals. For investors, the net Sharpe ratio of 0.43 indicates that FAC-based strategies remain profitable even after realistic implementation costs, suggesting potential value in incorporating this signal into quantitative investment processes.

Despite these promising results, several limitations warrant consideration. First, while our analysis controls for transaction costs, actual implementation may face additional frictions such as market impact and capacity constraints that could further erode returns. Second, the study period and market coverage may limit the generalizability of findings across different market regimes and international markets. Third, the economic mechanisms underlying the FAC anomaly require further investigation

to determine whether the premium represents compensation for systematic risk or market inefficiency.

Future research should explore several promising avenues. First, examining the performance of FAC across different market conditions and international markets would enhance our understanding of its robustness and universality. Second, investigating the economic channels through which fixed asset conservatism affects firm value and investor perceptions could provide deeper insights into the source of the return premium. Third, exploring potential interactions between FAC and other firm characteristics, particularly those related to investment efficiency and capital allocation decisions, may reveal additional nuances in the signal's predictive power. Finally, examining whether the FAC premium varies with institutional ownership, analyst coverage, or other measures of market efficiency could shed light on whether behavioral or risk-based explanations better account for the documented return predictability.

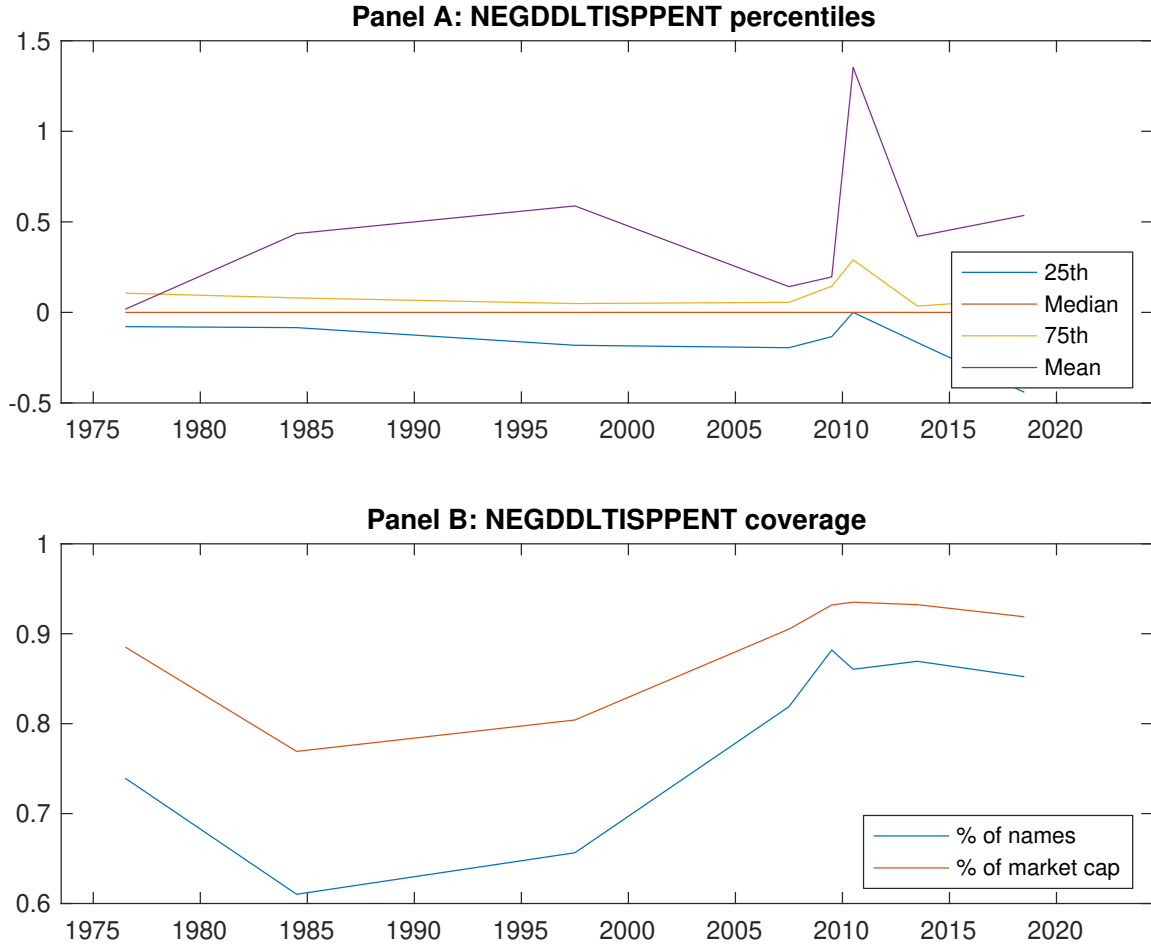


Figure 1: Times series of FAC percentiles and coverage.
This figure plots descriptive statistics for FAC. Panel A shows cross-sectional percentiles of FAC over the sample. Panel B plots the monthly coverage of FAC relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on FAC. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 197306 to 202406.

Panel A: Excess returns and alphas on FAC-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.57 [2.63]	0.68 [3.79]	0.64 [3.29]	0.69 [3.97]	0.85 [4.12]	0.27 [3.96]
α_{CAPM}	-0.17 [-2.98]	0.07 [1.46]	-0.01 [-0.22]	0.11 [2.05]	0.15 [2.77]	0.32 [4.69]
α_{FF3}	-0.21 [-3.74]	0.05 [1.07]	0.05 [0.84]	0.10 [1.92]	0.14 [2.66]	0.34 [5.11]
α_{FF4}	-0.17 [-3.02]	0.05 [1.16]	0.08 [1.51]	0.06 [1.10]	0.13 [2.54]	0.30 [4.44]
α_{FF5}	-0.17 [-3.10]	-0.03 [-0.64]	0.10 [1.83]	-0.00 [-0.05]	0.10 [1.89]	0.27 [4.03]
α_{FF6}	-0.14 [-2.64]	-0.02 [-0.42]	0.13 [2.26]	-0.03 [-0.51]	0.10 [1.89]	0.24 [3.65]
Panel B: Fama and French (2018) 6-factor model loadings for FAC-sorted portfolios						
β_{MKT}	1.11 [87.61]	0.98 [91.99]	0.95 [72.92]	0.95 [82.45]	1.06 [85.61]	-0.05 [-3.13]
β_{SMB}	0.12 [6.40]	-0.12 [-7.24]	-0.03 [-1.62]	-0.03 [-1.82]	0.15 [7.99]	0.03 [1.16]
β_{HML}	0.15 [6.12]	0.04 [2.11]	-0.10 [-4.01]	-0.07 [-3.34]	-0.06 [-2.44]	-0.21 [-6.93]
β_{RMW}	0.06 [2.31]	0.13 [6.37]	-0.03 [-1.10]	0.08 [3.77]	0.05 [1.92]	-0.01 [-0.35]
β_{CMA}	-0.22 [-5.95]	0.13 [4.07]	-0.17 [-4.45]	0.27 [8.03]	0.10 [2.87]	0.32 [7.14]
β_{UMD}	-0.04 [-3.08]	-0.02 [-1.49]	-0.04 [-2.98]	0.04 [3.22]	-0.00 [-0.12]	0.04 [2.41]
Panel C: Average number of firms (n) and market capitalization (me)						
n	712	491	1025	534	686	
me (\$10 ⁶)	1772	2743	2177	2666	1797	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the FAC strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 197306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.27 [3.96]	0.32 [4.69]	0.34 [5.11]	0.30 [4.44]	0.27 [4.03]	0.24 [3.65]
Quintile	NYSE	EW	0.15 [3.53]	0.17 [4.05]	0.17 [3.91]	0.16 [3.75]	0.15 [3.47]	0.15 [3.45]
Quintile	Name	VW	0.27 [3.81]	0.33 [4.63]	0.34 [4.85]	0.30 [4.19]	0.26 [3.69]	0.23 [3.34]
Quintile	Cap	VW	0.26 [3.98]	0.31 [4.79]	0.34 [5.18]	0.29 [4.37]	0.24 [3.80]	0.21 [3.31]
Decile	NYSE	VW	0.30 [3.20]	0.36 [3.77]	0.35 [3.62]	0.30 [3.10]	0.19 [2.01]	0.17 [1.80]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.21 [3.08]	0.25 [3.67]	0.27 [4.00]	0.25 [3.67]	0.21 [3.12]	0.19 [2.88]
Quintile	NYSE	EW	-0.12 [-1.95]					
Quintile	Name	VW	0.21 [2.94]	0.26 [3.68]	0.27 [3.84]	0.25 [3.50]	0.20 [2.91]	0.18 [2.66]
Quintile	Cap	VW	0.21 [3.19]	0.25 [3.83]	0.27 [4.14]	0.24 [3.74]	0.19 [2.98]	0.17 [2.68]
Decile	NYSE	VW	0.23 [2.44]	0.29 [3.05]	0.28 [2.89]	0.25 [2.63]	0.15 [1.59]	0.13 [1.40]

Table 3: Conditional sort on size and FAC

This table presents results for conditional double sorts on size and FAC. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on FAC. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high FAC and short stocks with low FAC. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 197306 to 202406.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	FAC Quintiles					FAC Strategies						
	(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}	
	(1)	0.69 [2.42]	0.89 [3.35]	0.95 [3.43]	0.92 [3.44]	0.77 [2.68]	0.09 [0.96]	0.11 [1.21]	0.09 [0.93]	0.07 [0.71]	0.04 [0.41]	0.03 [0.34]
	(2)	0.75 [2.79]	0.87 [3.48]	0.86 [3.42]	0.94 [3.82]	0.81 [3.16]	0.06 [0.71]	0.09 [1.13]	0.05 [0.65]	0.06 [0.82]	0.01 [0.13]	0.02 [0.31]
	(3)	0.80 [3.12]	0.81 [3.67]	0.79 [3.34]	0.83 [3.76]	0.91 [3.76]	0.11 [1.43]	0.15 [2.03]	0.15 [2.01]	0.11 [1.46]	0.16 [2.04]	0.13 [1.64]
	(4)	0.75 [3.25]	0.81 [3.84]	0.81 [3.67]	0.78 [3.83]	0.86 [3.86]	0.11 [1.44]	0.14 [1.80]	0.12 [1.60]	0.09 [1.19]	0.09 [1.19]	0.07 [0.93]
	(5)	0.50 [2.40]	0.62 [3.49]	0.61 [3.26]	0.57 [3.21]	0.87 [4.43]	0.37 [4.15]	0.41 [4.59]	0.43 [4.80]	0.35 [3.96]	0.32 [3.57]	0.27 [3.04]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	FAC Quintiles					FAC Quintiles						
	Average n					Average market capitalization (\$10 ⁶)						
	(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)		
	(1)	386	388	388	388	384	36	34	32	33	35	
	(2)	106	106	106	106	106	60	61	59	60	59	
	(3)	75	76	75	75	75	107	107	103	105	106	
	(4)	63	64	64	64	63	228	234	226	233	226	
(5)	58	58	58	58	58	1456	2127	1709	2145	1575		

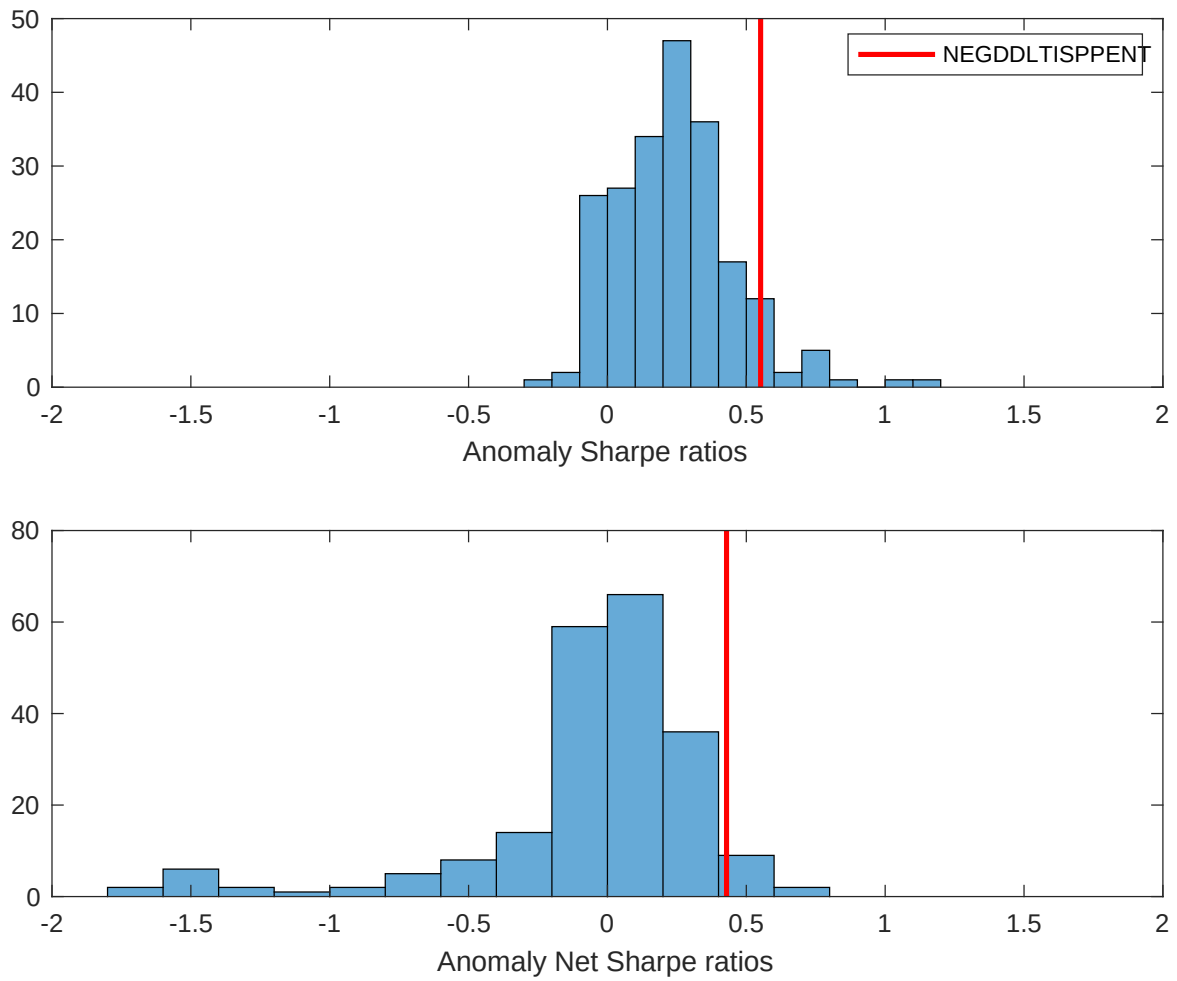


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the FAC with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

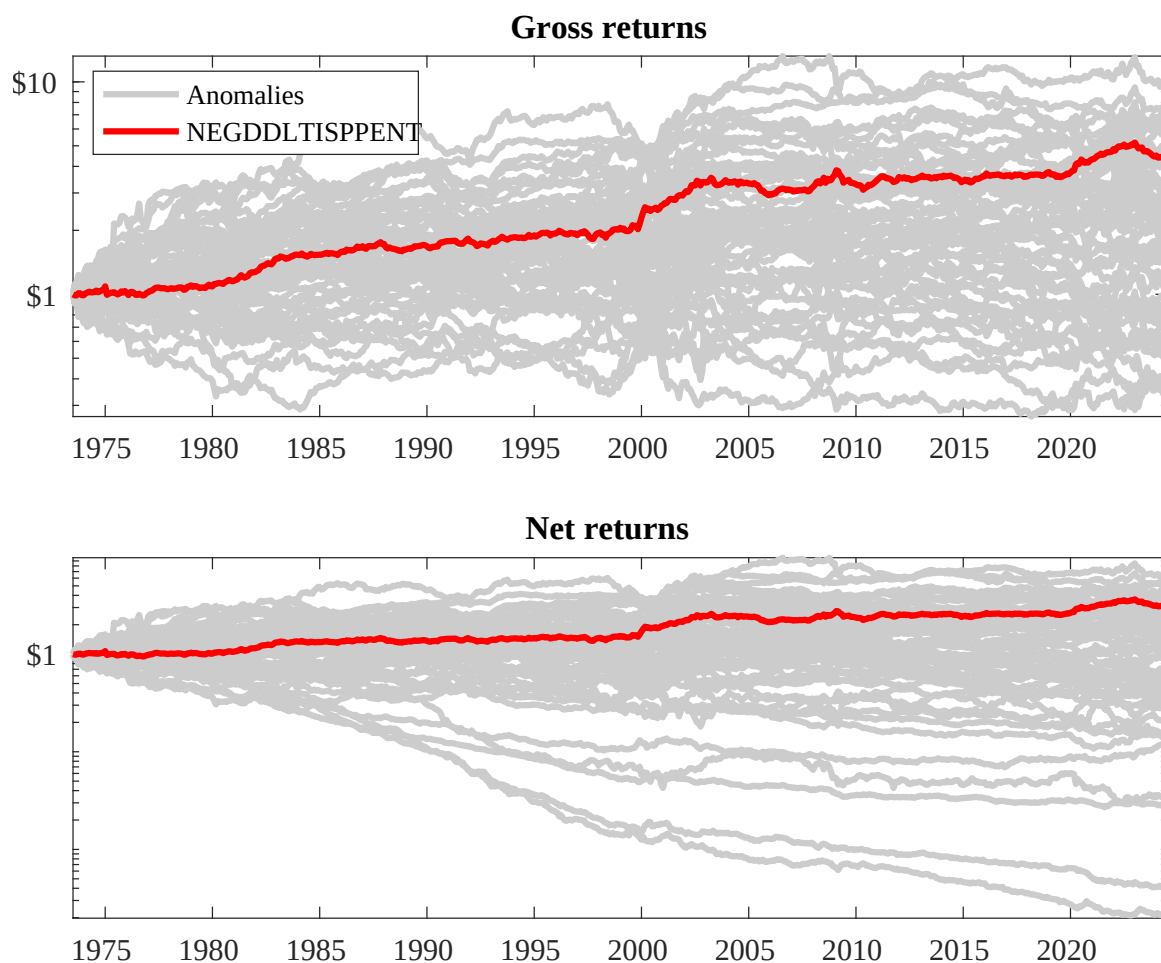
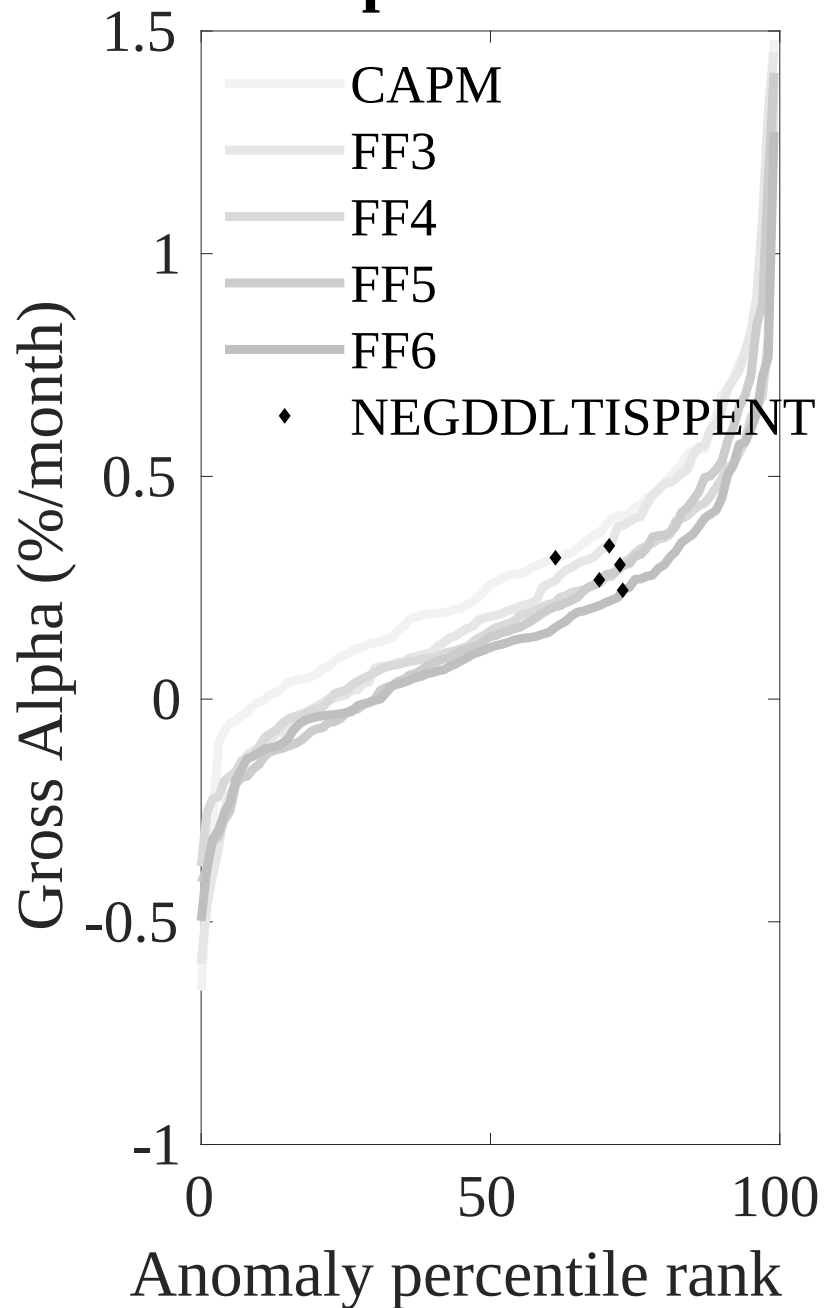


Figure 3: Dollar invested.

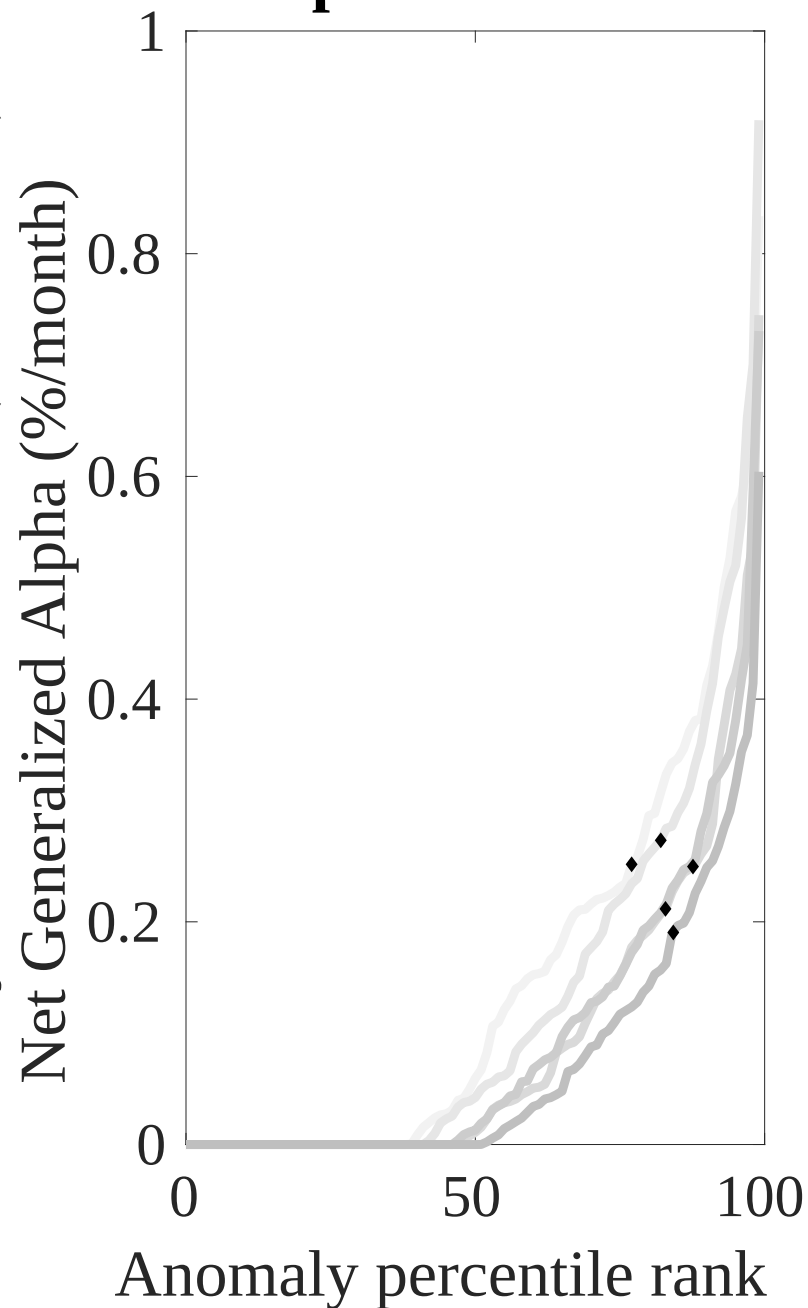
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the FAC trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

Gross Alpha Percentiles



Novy-Marx and Velikov (RFS, 2016)

Net Alpha Percentiles



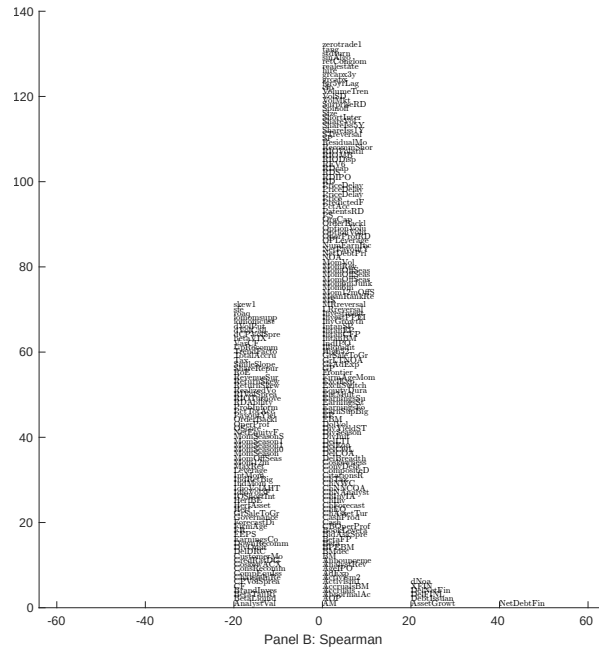
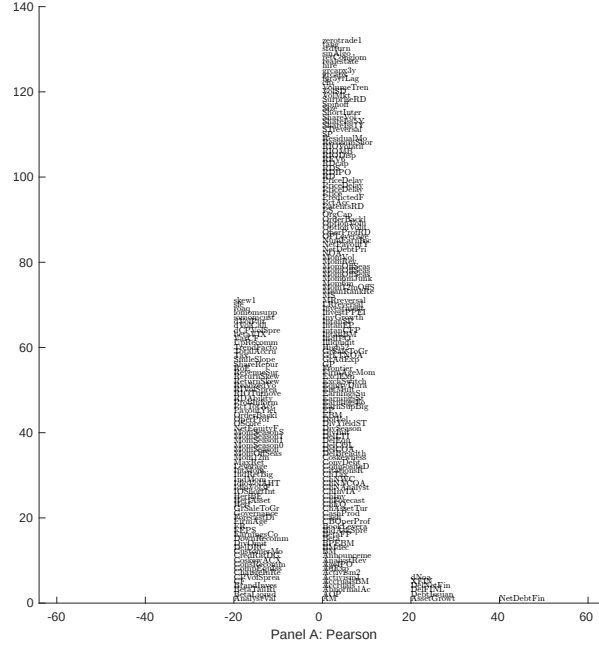


Figure 5: Distribution of correlations.

This figure plots a name histogram of correlations of 210 filtered anomaly signals with FAC. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

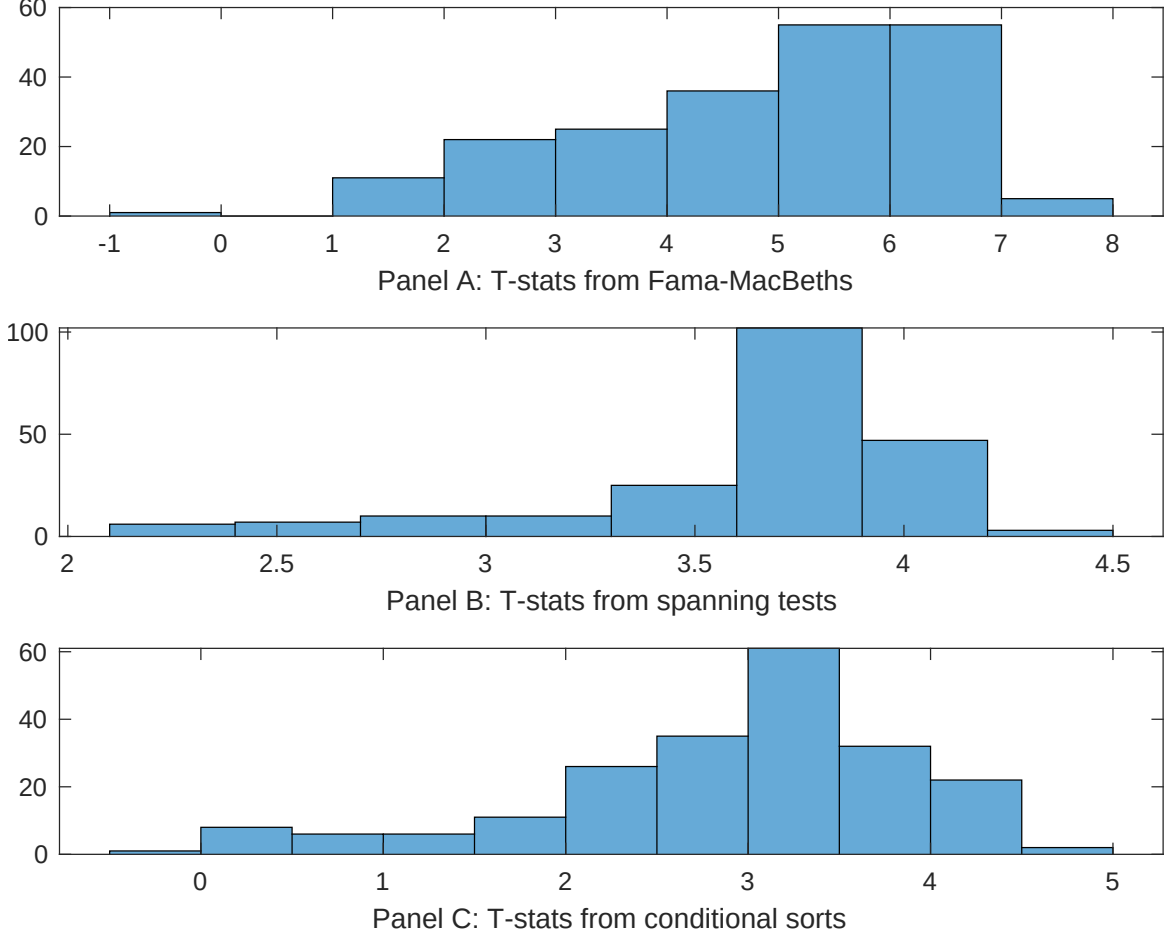


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of FAC conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{FAC} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{FAC} FAC_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{FAC,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on FAC. Stocks are finally grouped into five FAC portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted FAC trading strategies conditioned on each of the 210 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on FAC. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{FAC}FAC_{i,t} + \sum_{k=1}^s \beta_{X_k} X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Change in financial liabilities, Net debt financing, Net external financing, Asset growth, Inventory Growth, Accruals. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 197306 to 202406.

Intercept	0.13 [5.38]	0.13 [5.35]	0.14 [5.70]	0.14 [5.81]	0.14 [5.31]	0.12 [5.05]	0.14 [5.62]
FAC	0.34 [2.40]	0.45 [2.74]	0.52 [3.66]	0.32 [2.01]	0.80 [4.99]	0.84 [5.60]	0.16 [0.91]
Anomaly 1	0.17 [9.07]						-0.66 [-1.43]
Anomaly 2		0.20 [8.58]					0.91 [1.37]
Anomaly 3			0.19 [6.29]				0.10 [1.77]
Anomaly 4				0.11 [8.49]			0.71 [3.20]
Anomaly 5					0.41 [6.87]		0.60 [1.07]
Anomaly 6						0.14 [4.39]	0.52 [1.52]
# months	612	612	612	612	612	612	612
$\bar{R}^2(\%)$	0	0	1	0	0	0	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the FAC trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{FAC} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Change in financial liabilities, Net debt financing, Net external financing, Asset growth, Inventory Growth, Accruals. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 197306 to 202406.

Intercept	0.21 [3.23]	0.22 [3.31]	0.22 [3.26]	0.23 [3.46]	0.23 [3.51]	0.22 [3.31]	0.20 [3.02]
Anomaly 1	19.13 [5.14]						9.52 [1.91]
Anomaly 2		21.20 [5.94]					11.85 [2.42]
Anomaly 3			13.91 [4.19]				10.05 [2.85]
Anomaly 4				5.59 [1.44]			0.36 [0.09]
Anomaly 5					2.93 [1.12]		1.62 [0.63]
Anomaly 6						3.45 [1.30]	1.36 [0.52]
mkt	-4.15 [-2.74]	-4.55 [-3.03]	-2.68 [-1.69]	-4.60 [-2.99]	-4.66 [-3.02]	-4.34 [-2.79]	-2.88 [-1.81]
smb	0.76 [0.33]	0.84 [0.37]	7.19 [2.83]	2.32 [0.99]	3.04 [1.29]	3.38 [1.42]	4.33 [1.59]
hml	-19.24 [-6.58]	-19.65 [-6.78]	-18.95 [-6.41]	-20.75 [-6.97]	-20.58 [-6.92]	-19.61 [-6.44]	-18.04 [-6.04]
rmw	-2.75 [-0.92]	-3.35 [-1.13]	-9.34 [-2.61]	-1.00 [-0.33]	-0.60 [-0.20]	0.11 [0.03]	-8.51 [-2.30]
cma	26.33 [5.73]	27.23 [6.06]	23.65 [4.78]	25.94 [3.98]	30.10 [5.93]	31.59 [6.87]	17.64 [2.66]
umd	2.22 [1.44]	2.51 [1.66]	3.78 [2.50]	4.00 [2.60]	3.58 [2.30]	3.62 [2.34]	1.94 [1.25]
# months	612	612	612	612	612	612	612
$\bar{R}^2(\%)$	18	19	17	15	15	15	20

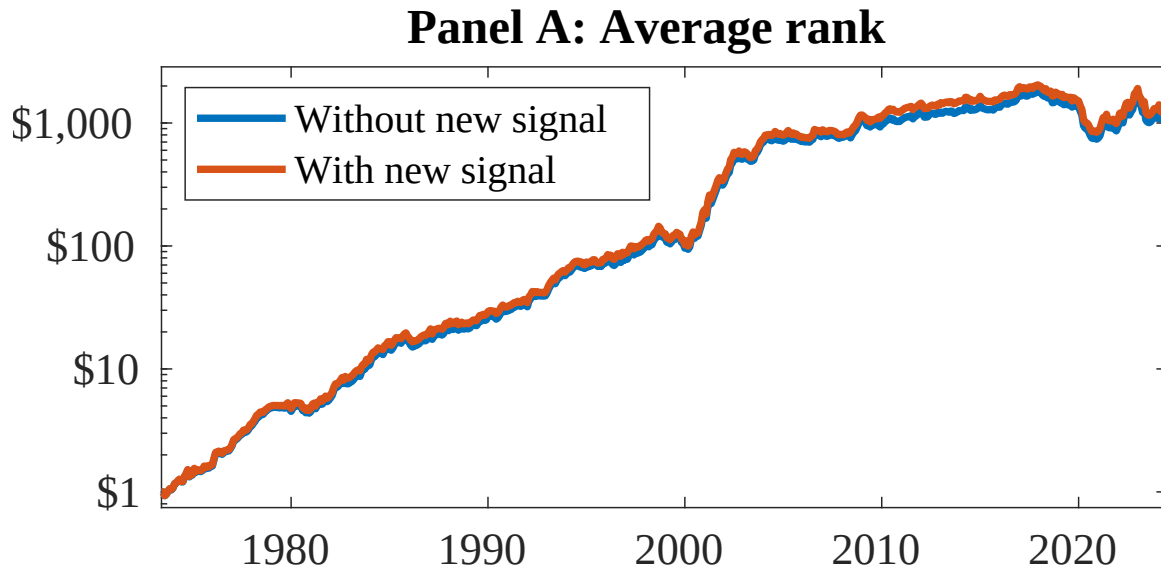


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 155 anomalies. The red solid lines indicate combination trading strategies that utilize the 155 anomalies as well as FAC. Panel A shows results using "Average rank" as the combination method. See Section 7 for details on the combination methods.

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